



AI Framework For Wall Damage Analysis And Cost Estimation: A Study For Sri Lankan Context

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A Thesis Submitted in Partial Fulfilment of the Requirement for the Degree of Bachelor of
Information and Communication Technology

August 2025

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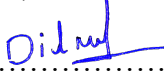
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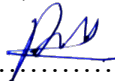
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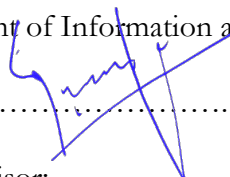
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UVA WELLASSA UNIVERSITY OF SRI LANKA

Abstract

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Building wall structure assessment and early repair are crucial for ensuring the buildings. Safety, durability, and economy, particularly in tropical nations like Sri Lanka, where climatic conditions intensify surface degradation. This paper presents a localized, image-based system that integrates damage classification, area localization, and cost estimation into a single, web-based platform. Harnessing deep learning architectures, ResNet-50 for typing classification and YOLOv8 for spatial detection, the model was trained on a custom-curated dataset representative of Sri Lankan wall conditions. Designed with real. Material and labor expenses in mind, the model not only predicts repair needs but also offers context-aware cost estimates and precise preventive recommendations through a Large Language Model. An interactive front end was designed to render it accessible to both technical and non-technical stakeholders. Validation results demonstrate promising detection accuracy and cost estimation validity despite recognized weaknesses in handling underrepresented types of damage and erratic economic inputs. The system offers a scalable, practical means to enhance conventional wall examination methods, enabling an active culture of infrastructure upkeep in low-resource contexts. It is also a replicable example for similar contexts around the Global South.

Keywords: cost estimation, detection of wall damage, deep learning, YOLOv8, ResNet-50, Sri Lanka, construction maintenance

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Chapter 1 : Introduction

1.1 Background and rationale

In the Sri Lankan construction industry, walls are the most critical element of a building since they carry the load of the building and protect it from climatic conditions. Due to the fact that Sri Lanka experiences a tropical climate, which is caused by high rain and humidity, walls are highly prone to all kinds of damage including cracking, spalling, water seepage, and biological growth. If not detected and fixed in a timely manner, these problems can affect the structural integrity, building occupant safety, and energy efficiency of buildings.

Better wall maintenance is thus an essential part of sustainable building. Conventionally, Sri Lankan wall damage observations have been based on visual observation by experienced engineers. These approaches are subjective, unreliable, and inadequate to assess inaccessible areas. The subsequent cost estimation process, after damage detection, is also manually performed, consuming time and with the possibility of errors. There are also no centralized digital systems for maintaining damage records or standard guidelines for inspection.

With the world advancing in Artificial Intelligence (AI) and computer vision, automated damage detection systems are becoming more feasible. Deep learning models such as ResNet and YOLOv8 have proven excellent crack detection, water stain detection, flaking, and other structural abnormality detection from images. Despite these advancements, there is minimal utilization of these types of technologies in Sri Lanka, primarily due to the lack of localized research and datasets.

This research is driven by the necessity of updating the existing wall inspection and cost estimation process. By creating an AI-based framework that is specific to the building conditions, materials, and economic conditions in Sri Lanka, this project seeks to deliver an extendable and automated solution that can aid engineers and decision-makers in infrastructure maintenance. The incorporation of damage detection and cost estimation through real-time image processing could greatly enhance the accuracy, efficiency, and reliability of building inspections within the local scenario.

1.2 Overview of the study

This research establishes an AI-driven framework that can automatically identify and categorize typical surface-level building wall damages in Sri Lanka, along with providing a localized cost of repair and prevention recommendation. Underpinned by state-of-the-art deep learning architectures like ResNet and YOLOv8, the framework examines user-submitted photos for various types of damage including cracks, water seepage, flaking paint, and missing parts.

To balance the lack of training data collected locally, the dataset was built through the merging of images collected manually and available material so that the dataset can serve as a replacement for Sri Lanka's environmental and building parameters. The cost estimation module can accept localized material and labor costs and can give practical and realistic estimates of the repair cost.

Furthermore, the system also includes a Large Language Model (LLM) Gemini to provide preventive recommendations from the recognized types of damage. This is an additional feature that makes the system a diagnostic, cost estimation, and knowledge-based long-term maintenance assistant tool.

By targeting wall surface damages that can be seen, this general framework will further enhance the accuracy, effectiveness, and decision-making in the construction and property management industries in Sri Lanka.

1.3 Problem Statement

Ensuring structural integrity and minimizing the cost of repair in the long run relies on the precise and timely identification of wall damage in structures. Traditional wall inspection and cost estimating procedures in Sri Lanka are manual, subjective, and time-consuming, which leads to inconsistent appraisal and underestimation of the required repairs. Although artificial intelligence (AI) has shown potential in automating damage detection and cost estimation globally, most of the models are trained using foreign datasets and are not concerned with the unique environmental, material, and economic conditions prevailing in Sri Lanka. These limitations undermine the effectiveness of AI in the Sri Lankan construction sector and result in inferior model performance upon deployment locally. Further, there is no comprehensive model integrating both functions in a single solution; AI systems tend to conduct either cost estimation or damage detection. Accordingly, there is a clear need for a context-aware, image-based AI system capable of identifying typical wall damage types and providing

accurate, localized repair cost estimation to facilitate decision-making in Sri Lankan construction practice.

1.3 Research Objectives

1. The goal is to gather and label a variety image dataset of common surface wall damages in Sri Lankan structures, with both manually captured photos and web-based images. This dataset focuses on salient types of damage, e.g., cracks, water infiltration, flaking paint, and missing areas, considering the construction materials common in the local area as well as the environment.
2. To develop and implement a classification and localization framework based on deep learning, employing architectures like ResNet for classifying damage types and YOLOv8 for localizing damages in specific regions, as well as cost estimation mapping.
3. To implement a Large Language Model (LLM) like Gemini to recommend preventive measures for each category of damaged types to assist users in knowing better maintenance strategies in order not to repeat future damage.
4. To create a friendly web application to offer users the functionality of posting wall pictures, to get automatic detection and cost estimates, and to get AI-generated prevention recommendations.
5. To ascertain the accuracy, reliability, and usability of the end-to-end system, the effectiveness of it has to be tested within the technological, economic, and environmental conditions of Sri Lanka's building maintenance industry.

1.4 Research Questions

Below are the research questions that will direct this study:

1. What are common types of surface level wall damages in Sri Lankan buildings, and how are they best categorized with deep learning-based image analysis?
2. How do traditional cost estimation protocols differ from AI-based estimation models on the basis of their accuracy, adaptability, and productivity, especially under local economic contexts?
3. How do composite ResNet and YOLOv8 models assist in classifying types of wall damage and locating damaged areas to estimate repair expense?

4. What is the potential of Large Language Models (LLMs) for individualized preventive maintenance suggestions according to recognized types of wall damage and building patterns?
5. How do AI-powered technology solutions tailored to the challenges of the Sri Lankan construction sector solve region-specific issues of data scarcity, heterogeneity of material types, and heterogeneity of environmental conditions?

The purpose of such research questions is to investigate a total integration of computer vision and natural language processing as an interactive, user-focused medium for damage detection, cost estimating, and preventive maintenance guidance in Sri Lankan construction.

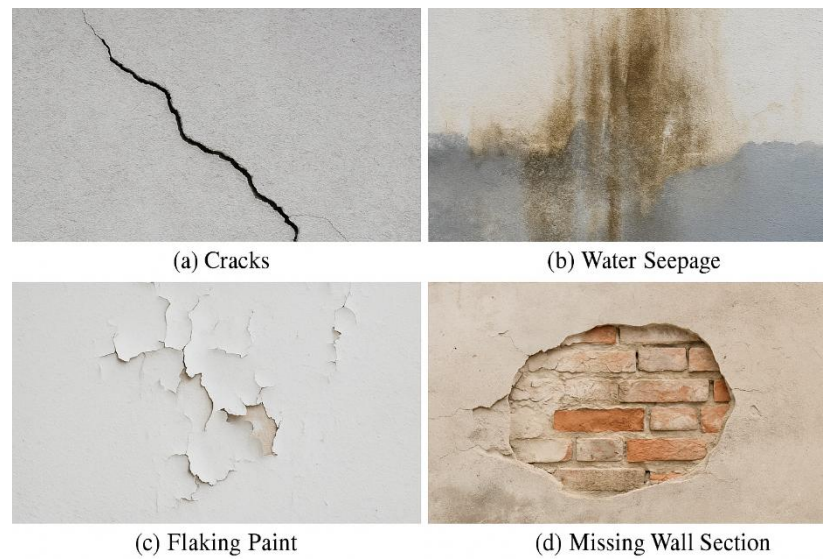


Figure 1.1: Common wall types in Sri Lankan Construction

1.5 Scope of the Study

This material addresses visible surface imperfections such as cracks, seepage of water, flaking paint, the absence of wall sections, and weathering impacts, which are prevalent in Sri Lankan structures. It can be used in urban and semi-urban Sri Lanka, where materials such as concrete, brick, cement blocks, and composite wall types are common.

Technologically, the framework involves a hybrid deep learning pipeline based on ResNet for classification and YOLOv8 for localization. A cost estimation module uses local labor costs and the price of materials to provide location-specific repair quotes, limited to surface-level work such as plastering, repainting, and waterproofing. A LLM element is also incorporated to give preventive

advice depending on the severity and nature of damage encountered, helping users to know how to avoid similar occurrences in the future.

Research does not cover full-scale structural rehabilitation, internal or non-visible wall defects, and engineering evaluation with load-bearing analysis. Cost estimation only covers minor to moderate repair and does not cover large-scale construction or new works.

This well-specified scope ensures a practical, cost-effective, and technology-driven solution designed to allow building owners, contractors, and maintenance experts to make informed decisions through an intuitive, web-based application.

1.7 Significance of Study

This study provides useful contributions to the academic world and also practical applications in the infrastructure maintenance and artificial intelligence domain. Academia-wise, it takes a step forward in construction-oriented computer vision since it demonstrates the successful adaptation and application of deep learning architectures vs. ResNet and YOLOv8 to address region-specific issues of developing countries like Sri Lanka. Furthermore, the integration of a Large Language Model (LLM) to provide preventive recommendations introduces a new aspect to AI-powered maintenance systems with inter disciplinary in civil engineering, data science, and natural language processing. In practice, the research provides an effective, low-cost, and easy-to-apply system for automatic detection of damage on wall surfaces, estimation of repairs, and recommendation of preventive suggestions.

The system can be a valuable resource in the hands of government agencies that are tasked with maintaining public infrastructure, enabling smarter budgeting and quicker response. Similarly, private developers, construction companies, and property managers can enjoy lesser manual inspection times, greater accuracy in planning repairs, and increased long-term asset durability. Through utilizing AI for otherwise low-technology and labor-intensive operations, the work causes digital innovation in Sri Lanka's construction industry. It enables active, evidence-based infrastructure management procedures, in addition to displaying a replicable model which can be exported to other low-resource, tropical, or monsoon-based regions of the world

1.8 Limitations of the Study

Although this study proposes a new and innovative AI-based wall damage detection, estimation, and repair cost system, the author recognizes the following limitations:

Static Cost Estimation: The cost estimation feature is founded on a set of material and manpower costs which has been ascertained earlier, and which is aggregated as development proceeds. The system thus lacks real time variations in market changes, and this may also cause the accuracy of estimates to slip whenever there are variations in the economy.

Damage Identification Only Surface-level: The system is only capable of detecting visible damage such as cracks, peeling paint on walls on the one hand, or invisible damage such as micro cracks on the other. It cannot evaluate internal depth and severity of the damage and structural effects which would be in special equipment range or engineering inspections.

Environmental Conditions Sensitivity: The differences in quality in the images that depend on the light conditions, the existence of shadows, or even the differences according to the device used could affect the performance of the detection model when the users provide such images that have a different light condition.

Generalization Limitations of Material Specificity: Although the model is trained in regard to picture of standard material surfaces in Sri Lanka consisting of concrete, brick, and cement block, its classification performance might be worse when its predictions are used on rare, unusual, or underrepresented types of wall materials.

Exclusion of Structural Load Analysis: The analysis does not cover any structural load or surface and cosmetic damage only. It does not include an evaluation of structural load-carrying capacity, and it does not constitute a study covering the effect which the observed destructions are going to have on the integrity of a building.

These shortcomings imply a set of directions that would be needed to work on in the future, i.e., to add on-time pricing APIs, sophisticated imagining tools (e.g., thermal or 3D scans), and structural health monitoring modules to extend the breadth and precision of the framework.

1.9 Structure of the Thesis Document

This paper is divided into eight substantial chapters that are arranged in such a way so that they had a logical flow presenting the research backbone, the development process, and results.

- Chapter 1: Introduction - Presents the background of the research, rationale, objectives, problem statement, research questions, scope, as well as significance associated with the study.
- Chapter 2: Literature Review - Reviews existing literature on AI- based wall damage detection and cost estimation services, gaps in the literature and requirement of locally applicable solutions.
- Chapter 3: Materials and methods - This section provides the research design and dataset collection, annotation procedure, model training procedure, and evaluation techniques.
- Chapter 4: Tools and Technologies - Outlines the technical equipment's, software libraries, programming environments, hardware requirements, and the structure of the system to be used to develop and implement the proposed framework.
- Chapter 5: System Development and Implementation - Describes system development and implementation, step-by-step details include: model integration, development of web application and internal testing.
- Chapter 6: Results and Discussion - Puts the findings of the evaluation of the damage detecting and cost estimators and explains their applicability to real-life.
- Chapter 7: Conclusion and Recommendations Summarize the most significant details of the research, reflects the purpose of the research and gives practical recommendations about adopting and scaling.
- Chapter 8: Future Work- Presents possible opportunities to strengthen the system and increase their application through expansion of the datasets, real-time cost modification, and larger areas of use within infrastructure maintenance.

Chapter 2 : Literature Review

2.1 Introduction

Surveillance of the structural stability of a building and its economic sustainability relies a great deal on the timely and accurate identification of the damage in the wall. The construction industry is also under increased pressure to deploy the most advanced, scalable, and ingenious solutions to the infrastructure maintenance since there is growing urbanization in Sri Lanka. The possible use of the technologies of artificial intelligence (AI) in the system of computer vision and deep learning to automate the process of damage detection and help model future costs has attracted the attention of the global community [1]. In this chapter, current research papers talk about the introduction of AI to the construction industry, how fit current models are to the Sri Lankan context, and international and regional literature that is relevant to the topic. The proposed study is aimed at the deployment of a web-based artificial intelligence system that utilizes deep learning models, including ResNet and Enhanced YOLO, to perform damage and area detection, together with a local cost estimation module implemented by Sri Lankan material, and labors rates.”

2.2 Scope of the Review

The review discussion will cover the global trends in artificial intelligence-based wall damage detection and cost estimation of construction works, primarily through the use of images. Convolutional Neural Networks (CNNs), Fully Convolutional Networks (FCNs) and hybrid AI models are mentioned regarding deep learning. Realizing economic, technological and environmental constraints, the chapter also concentrates on the environment of Sri Lanka. Such problems as high humidity and inconsistent building, as well as the use of datasets with an insufficient degree of transferring the global models that have already been learned are considered with particular attention [2], [3]. The contextual restrictions also portray the importance of developing domain-specific datasets and models of AI trained in Sri Lankan walls to improve the generalization and applicability.

2.3 Existing Applications and Themes

The application of CNNs has become common in automated detection of wall defects such as water seepage, corrosion, cracking, spalling, and deterioration of paint. When it comes to classification of images in case of surface level structural damage, architectures like the VGGNet, ResNet, and MobileNet have been found to be very reliable [4] [5] [6]. The additional pixel-wise accuracy is obtained with segmentation networks such as U-Net and DeepLabV3+ [7] [8] In a bid to be more

robust in different situations, recent research has incorporated residual blocks, pixel-local weighting, and attention operations [9] Applications are not limited to infrastructure however, they have been used to monitor relevant structures such as bridges under strain or to preserve a culture by preserving its heritage structures [10] Regardless, the available datasets to train such models are mostly about East Asian or Western countries, which are subjected to different types of weather-related or material-related damage compared to Sri Lanka [11]

2.4 AI in Construction Cost Estimation

Traditional methods of cost estimation such as expert judgment estimation, unit-based estimation and calculation methods are ineffective due to their subjectivity [12]. Examples of AI-based methods with demonstrated enhanced scalability, flexibility, and real-time accuracy include Artificial Neural Networks (ANNs), Support Vector Regression (SVR), Random Forest (RF), and fuzzy Analytic Hierarchy Process (fuzzy-AHP) to create dimensional data and predict associated costs according to the identified complexity of construction tasks [13] Similar to the models mentioned, though, most of these models utilize data that reflects the global construction industry average. These models fail to explain the oscillations of the labor rates in Sri Lanka, the costs of the materials, and the logistics issue until tempered with local data [15].

2.5 Understanding Wall Damage Types and Classification Needs

To be able to effectively plan rehabilitation, one needs to classify types of damage, including surface cracks, moss growth, flaking paint, and missing pieces of the wall. Multi-label classification and semantic segmentation [16] has been used to make interpretations of patterns of compound or co-occurring damage. These methods enhance AI capabilities to match what they see on the screen with what repairs need to be done and what the costs estimate will look like. Sri Lanka also has different damage profiles, as regions are humid, so there is black mold, areas along the coast have salt-induced corrosion, and inadequate painting surfaces cause flaking paint, which is not well represented in available datasets, this will require having a local image data set that can be used to improve classification performance.



Figure 2.1: Crack in the wallⁱ



Figure 2.2: Water damage in the wall



Figure 2.3: Flaking Paint in the wall



Figure 2.4: Missing piece in the wall

2.5 Localized AI Solution and the Sri Lankan Context

Nevertheless, the Sri Lankan situation is far behind in adopting these tools, despite other research globally conferring that AI has the potential to complete such tasks in diagnostics that related to infrastructure. Among the main obstacles, there are poor digital infrastructure, the deficiency of AI knowledge in the building industry, and the absence of well-chosen local environmental data [18]. Furthermore, models trained in a different environmental setting have many cases of misclassification or poor performance when used in the local area. Such as, the discoloration by the tropical algae could be categorized improperly as the mold or the weathering could not be noticed on the laterite bricks [25]. Also, the current solutions tend to halt at the stage of detecting the damages and do not go as far as estimating the costs with consideration of local prices and labor conditions. This loophole underlines the importance of having an integrated system that is Sri Lanka-specific connecting visual diagnosis and proper estimation of repair. [19]

2.7 Market Trends and Industry Landscape

2.7.1 Market Characteristics and Emerging Trends

2.7.1.1 Characteristics

Sri Lanka is experiencing the rapid process of urbanization, and it is also required to have even more maintenance of its infrastructure, which promotes the growth and development of the construction industry.

Crucial attributes of the market are as follows:

1. **Urbanization:** The need of getting the real damage detection and costs estimation is also growing and, in particular, in the city planning course as new buildings and reconstruction need more buildings and enough constructions.
2. **Technological Adoption:** Although the wider use of AI in the construction industry remains underdeveloped, it is becoming more popular to use the technology to address the rightful assessment of damages and cost estimates.
3. **Labor Shortages:** To automate damage quantification and not have to hire skilled labor, one will require AI solutions due to scarcity of such skilled labor.
4. **Cost control:** The cost control procedures are to be exact in order to remain competitive within the construction business. AI simplifies things and makes them more precise and cheaper.

5. Sustainability: With the outpouring of green buildings in Sri Lanka, the very idea of creating a sustainable building using AI-based solutions is within the representative demand.

2.7.1.2 Trends:

- **AI adoption:** Predictive maintenance, material optimization and detection of damaged assets are some of the areas that AI is being more widely applied.
- **Image Recognition:** It is being observed that CNNs and localized datasets of images are being applied in damage detection, thus making the Sri Lankan structures more precise.
- **Data-Driven Decisions:** AI enables more informed decisions to be made in terms of estimating the cost and determining the extent of damage.
- **Cloud Integration:** To track all activity and collaborate in real-time, a mix of cloud technology and Ai-based technologies are used.
- **Cost-Effectiveness:** AI applications make construction projects more cost-effective as they offer faster and more accurate scans of the damage.

Such trends highlight the heightened importance of AI in efficiency and cost-cutting solutions as well as maintaining a more satisfactory construction result in Sri Lanka.

2.7.2 Industry Analysis

Getting a loan to buy a house or an apartment becomes easier with the help of AI-driven technologies that construction sector has adopted to make cost estimation processes much more efficient and accurate. The solutions provide estimation of the costs on the basis of the labor data and the size of the project, material consumption and past performance with the help of the hybrid models, machine learning, and neural networks. Computer vision and BIM (Building Information Modeling) are also now used with AI to automate the cost forecasting and quantity takeoff of projects. Even though other parts of the world are at different stages of having these technologies embraced within the mainstream building, Sri Lanka is still in its infant stages of adopting these technologies to the mainstream building practices of the country.

Table 2.-1: Summary of Current AI Based Tools and Models for Construction Cost Estimation

Tool/Model	Developer/Institution	Core Technology Used	Key Features	Availability
Autodesk Construction Cloud	Autodesk	ML, Predictive Analytics	Forecasting, cost modeling, integration with BIM	Commercial
Candy (CCS)	RIB CCS	AI-enhanced Estimating Engine	Tendering, project control, and cost comparison	Commercial
Smart Estimator	Causeway	Rule-based + ML Integration	Quantity takeoff, cost libraries, cloud collaboration	Commercial
unplan	nPlan (UK)	Deep Learning	Forecasts cost and schedule risks using historical data	Commercial (Enterprise)
OneClickLCA	Bionova Ltd.	AI-Driven LCA	Life-cycle cost & carbon estimation	Commercial
AI-Cost Estimator Plugin for BIM	Various (Integrated in BIM tools)	AI+BIM	Real-time estimate updates from 3D models	Varies (Commercial/Plugin)

2.8 Summary of Key Findings

According to the review, CNNs and other AI techniques have been successfully applied globally for wall damage detection and construction cost forecasting. Most solutions, though, are either region-specific or rely on datasets that don't accurately reflect the circumstances in Sri Lanka. The main drawbacks are the lack of integrated cost estimation that is in line with local markets, misclassification brought on by tropical damage patterns, and dataset incompatibility.

2.9 Identified Research Gaps

There is an evident research gap to develop an optimized system, using AI, that is specially adapted to the context of Sri Lanka and combines localized estimation of costs with the detection of damage. Local annotations on images are hardly ever used to train systems and process and perform such tasks on their own. The requirement of a cost model with the use of current market-rate and a publicly available image dataset that is specific to Sri Lanka does not have any existing literature as of now.

Table 2.-2: A Summary of previous studies on Damage detection and cost estimation

Ref no.	Author(s)	Year	Focus Area	Methodology	Key Findings	Limitations
1	K.Chen et al.	2024	AI in infrastructure construction	Critical review	Comprehensive analysis of AI applications	Limited to theoretical discussion
2	K. Tijanić Štrok	2024	Cost estimation in high-rise construction	Mathematical modeling	Improved cost estimation accuracy	Region-specific applicability
3	P. Dinesh Asanka et al.	2014	AI in Sri Lankan software industry	Survey	AI adoption in Sri Lanka	Outdated data
4	Y. Alzubi et al.	2024	Cost estimation of residential buildings	Machine learning	ML enhanced estimation accuracy	Middle East context
5	T. Narbaev et al.	2024	Cost estimate reliability	ML study	Improved reliability using ML	Limited dataset
6	A. H. Ali et al.	2024	AI in cost estimation	Exploratory analysis	AI's positive impact on cost practices	Lacks quantitative validation
7	E. Abu-Mahfouz et al.	2025	Hybrid AI models	Fuzzy-AHP + ANN	Enhanced cost estimation precision	Complex implementation
8	S. Teng et al.	2022	Structural damage detection	CNN	Effective bridge damage	Focused on bridges only

					identificati on	
9	L. Chen et al	2023	Bridge damage recognition	CNNs	Detected multiple bridge damage types	HSR RC bridges only
10	B. Duran et al.	2025	Damage ID via strain	Custom CNN	Robust strain-based damage detection	Requires sensors
11	M. Mishra et al.	2024	Cultural heritage monitoring	Visual inspection (AI)	AI aids structural health monitoring (SHM)	Heritage-specific
12	V. M. di Mucci et al.	2024	Bridge health management	AI models	Streamlined bridge SHM	Existing bridges focus
13	H. D. Nguyen et al.	2025	Seismic damage prediction	ML models	Predicted RC wall damage states	Limited to seismic impact
14	R. Ali et al.	2021	Crack segmentation	FCN + Residual learning	Effective pixel-level segmentation	Needs high-quality images
15	H. Bae and Y. K. An	2023	Crack quantification	Computer vision	Statistical crack analysis	Surface cracks only
16	E. Harirchian et al.	2021	Building safety evaluation	Soft computing	Rapid visual damage classification	Based on literature
17	K. Hacıfendioğlu et al.	2024	Masonry damage estimation	Deep learning	DL for post-earthquake damage segmentation	Limited to masonry
18	Wickramathilake and Hemachandra	2025	School building durability in Sri Lanka	Field study	Listed defects and solutions in schools	School context only

19	N. Fernando et al.	2024	Bridge cost estimation in Sri Lanka	ANN	Early-stage bridge cost prediction	Sri Lanka-specific
20	Chen Y,Wu W,Li J	2024	Wall crack detection using advanced object detection	YOLOv5 + attention modules; ResNet style backbone architecture	MAP improved ~2-3% over baseline; better accuracy on challenging textures	No cost estimation, dataset not local to Sri Lanka
21	Piyathilaka L, Preethichandra D, izhar U, Kahandawa G	2020	Concrete crack Localization	YOLACT (YOLO + mask-based segmentation) with ResNet-50/101 backbone)	ResNet-101 improved segmentation; ResNet-50 faster real-time FPS	Small dataset; may struggle on diverse wall conditions
22	Pan X, Yang T	2021	Damage Detection+ cost estimation	YOLO-v2 for area detection + ResNet-style classifier + cost model	~98% train precision, cost estimated for rapid deployment	Not localized to Sri Lanka; cost module basic

2.10 Conclusion

This chapter investigated the regional and international applications in the geographical extent of wall damage estimation and cost projection using AI. When it was used on construction cases in Sri Lanka, it showed the effectiveness of CNN based models and their constraints. The literature supports very much an idea to come up with localized solution with ability to estimate cost that is sufficient to meet the economic situation of Sri Lanka besides detecting accurately any damage. The steps followed in developing such a system are given in the next chapter.

Chapter 3 : Materials and Methods

3.1 Research Design

The given research will assume a mixed-methods experimental design to develop and confirm an AI-based solution to automatic wall damage identification and cost-artificial intelligence estimator designed specifically to serve the construction industry in Sri Lanka. The research process will follow a simple and repeated process which can be classified as the following key stages:

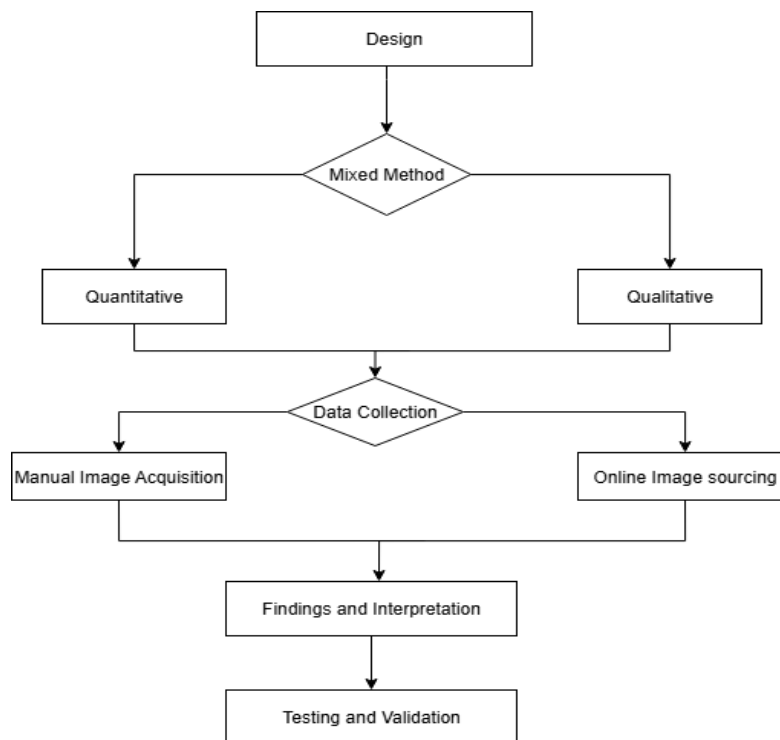


Figure 3.1: Research Design Diagram

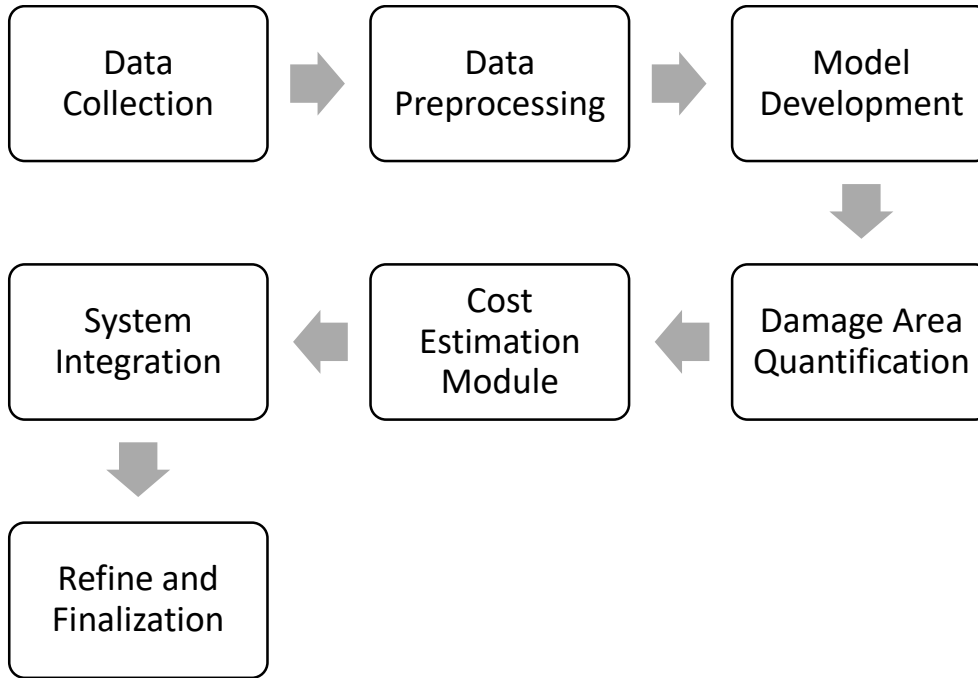


Figure 3.2: AI-Based Wall Damage Detection and Cost Estimation Workflow

Phase 1: Data Collection

The proposed AI framework is based on data collection. Various and representative image set was created in two main ways:

- **Manual Image Acquisition:** The field visits were arranged at different select places around urban and semi-urban localities of Sri Lanka. The high-resolution images of structural damage were captured using smartphones to ensure they form a real part of the environment.
- **Online Image Sourcing:** Additional sources of information were found on open platforms. The collection of images was selected based on how relevant it is to regional building types, how good the resolution is, and how ethical usages can be found.

Phase 2: Data Preprocessing

The below preprocessing requirements were undertaken to enhance the quality and usability of the dataset in the training of the model:

- **Resizing and Normalization:** All the images were transformed to standardized resolution to apply it to deep learning models.

- Data Augmentation: Methods such as rotation, flipping, cropping, and brightness adjustment were used to subsequently undergo artificial diversification of the data set in order to help fortify model resistance.
- Annotation and Labeling: A manual labelling of the images based on the type of damage occurred (e.g., cracks, water leaks, peeling) and supplemented by the metadata about the location, material type, the estimated repair costs were carried out.

```

1 import os
2 import logging
3
4 # Setup logging
5 logging.basicConfig(filename='../preprocess_errors.log', level=logging.ERROR, format='%s\n - %s\n')
6
7 # Create temporary directory for ResNet-50 resized images
8 resized_dir = '../data/resized'
9 os.makedirs(os.path.join(resized_dir, 'train'), exist_ok=True)
10 os.makedirs(os.path.join(resized_dir, 'val'), exist_ok=True)
11
12 # Recursively copy and resize images
13 for split in ['train', 'val']:
14     df = pd.read_csv('../split_metadata.csv')
15     for i, row in df.iterrows():
16         img_path = row['filepath']
17         if not os.path.exists(img_path):
18             print(f'Warning: Image not found: {img_path}')
19             logging.error(f'Image not found: {img_path}')
20             continue
21         if not img_path.startswith('../data/processed'):
22             print(f'Error: {img_path} points to invalid path (expected data/processed): {img_path}')
23             logging.error(f'Invalid path in metadata: {img_path}')
24             continue
25         filename = os.path.basename(img_path)
26         damage_type = row['damage_type']
27         resized_path = os.path.join(resized_dir, split, damage_type, filename.replace('/', '_'))
28         os.makedirs(os.path.dirname(resized_path), exist_ok=True)
29         try:
30             shutil.copy(img_path, resized_path)
31             img = image.open(resized_path)
32             img = img.resize((224, 224))
33             img.save(resized_path)
34         except Exception as e:
35             print(f'Error processing {img_path}: {e}')
36             logging.error(f'Error processing {img_path}: {e}')
37     print(f'Resized {split} images in data/resized.')
38
39 # TODO: Prepare dataset
40 os.makedirs('../yolo_dataset/images/train', exist_ok=True)
41 os.makedirs('../yolo_dataset/images/val', exist_ok=True)
42 os.makedirs('../yolo_dataset/labels/train', exist_ok=True)
43 os.makedirs('../yolo_dataset/labels/val', exist_ok=True)
44
45 train_df = pd.read_csv('../train_metadata.csv')
46 val_df = pd.read_csv('../val_metadata.csv')
47
48 for split, df in [('train', train_df), ('val', val_df)]:
49     for i, row in df.iterrows():
50         src_img = row['filepath']
51         filename = os.path.basename(src_img)
52         damage_type = row['damage_type']
53         dst_img = os.path.join('../yolo_dataset/images/' + split, filename.replace('/', '_'))
54         # Copy image
55         try:
56             if os.path.exists(src_img) and src_img != dst_img:
57                 shutil.copy(src_img, dst_img)
58             else:
59                 print(f'Warning: {src_img} does not exist or is the same as {dst_img}')
60                 logging.warning(f'{src_img} does not exist or is the same as {dst_img}')
61         except Exception as e:
62             print(f'Error copying {src_img} to {dst_img}: {e}')
63             logging.error(f'Error copying {src_img} to {dst_img}: {e}')

```

Figure 3.3: Preprocessing Procedure for Wall Damage Detection Dataset

The following image describes the preprocessing pipeline of the raw image dataset, where it includes resizing and normalization, augmentation (e.g. flipping upside-down, rotating), and labeling. Such measures enhance data consistency and heterogeneity in order to ramp up efficiency of the model during the training process

	A	B	C
1	filename	damage_type	filepath
2	crack_damages_001.jpg	crack_damages	../data/raw/crack_damages/crack_damages_001.jpg
3	crack_damages_002.jpg	crack_damages	../data/raw/crack_damages/crack_damages_002.jpg
4	crack_damages_003.jpg	crack_damages	../data/raw/crack_damages/crack_damages_003.jpg
5	crack_damages_004.jpg	crack_damages	../data/raw/crack_damages/crack_damages_004.jpg
6	crack_damages_005.jpg	crack_damages	../data/raw/crack_damages/crack_damages_005.jpg
7	crack_damages_006.jpg	crack_damages	../data/raw/crack_damages/crack_damages_006.jpg
8	crack_damages_007.jpg	crack_damages	../data/raw/crack_damages/crack_damages_007.jpg
9	crack_damages_008.jpg	crack_damages	../data/raw/crack_damages/crack_damages_008.jpg
10	crack_damages_009.jpg	crack_damages	../data/raw/crack_damages/crack_damages_009.jpg
11	crack_damages_010.jpg	crack_damages	../data/raw/crack_damages/crack_damages_010.jpg
12	crack_damages_011.jpg	crack_damages	../data/raw/crack_damages/crack_damages_011.jpg
13	crack_damages_012.jpg	crack_damages	../data/raw/crack_damages/crack_damages_012.jpg
14	crack_damages_013.jpg	crack_damages	../data/raw/crack_damages/crack_damages_013.jpg
15	crack_damages_014.jpg	crack_damages	../data/raw/crack_damages/crack_damages_014.jpg
16	crack_damages_015.jpg	crack_damages	../data/raw/crack_damages/crack_damages_015.jpg
17	crack_damages_016.jpg	crack_damages	../data/raw/crack_damages/crack_damages_016.jpg
18	crack_damages_017.jpg	crack_damages	../data/raw/crack_damages/crack_damages_017.jpg
19	crack_damages_018.jpg	crack_damages	../data/raw/crack_damages/crack_damages_018.jpg
20	crack_damages_019.jpg	crack_damages	../data/raw/crack_damages/crack_damages_019.jpg
	image_metadata	+	

Figure 3.4: Metadata Sheet for Wall Damage Classification Dataset

This picture presents a sample metadata table with annotated data for each picture, i.e., damage type, metadata helps in supervised learning calibration and cost estimation.

Phase 3: Model Development

This stage is devoted to the training and testing of deep learning models in the case of object detection and classification:

- To spot and identify damages areas on wall pictures in the form of bounding boxes, YOLOv8 (You only look once) was used.
- The fine-grained classification of the identified damage types was performed by means of ResNet-50 (Residual Neural Network).
- The data was divided into training (70%), validation (15%), and testing (15%). The aim of doing so was to test the performance of the models.

```
32 for damage_type in ["crack_damages", "flaking_paint", "missing_piece", "water_damage"]:  
33     os.makedirs(os.path.join(processed_dir, split, damage_type), exist_ok=True)  
34  
35 # Copy images and update filepaths  
36 for split, df in [("train", train_df), ("val", val_df)]:  
37     new_filepaths = []  
38     for _, row in df.iterrows():  
39         src = row["filepath"]  
40         filename = os.path.basename(src)  
41         damage_type = row["damage_type"]  
42         dst = os.path.join(processed_dir, split, damage_type, filename).replace("\\", "/")  
43         try:  
44             if os.path.exists(src):  
45                 shutil.copy(src, dst)  
46                 new_filepaths.append(dst)  
47             else:  
48                 print(f"Warning: Source file not found: {src}")  
49                 logging.error(f"Source file not found: {src}")  
50                 new_filepaths.append(dst) # keep dst path  
51         except Exception as e:  
52             print(f"Error copying {src} to {dst}: {e}")  
53             logging.error(f"Error copying {src} to {dst}: {e}")  
54             new_filepaths.append(dst)  
55     df["filepath"] = new_filepaths  
56  
57 # Save metadata  
58 train_df.to_csv("../train_metadata.csv", index=False)  
59 val_df.to_csv("../val_metadata.csv", index=False)  
60 print(f"Training images: {len(train_df)}, Validation images: {len(val_df)}")
```

Figure 3.5: Train Validation Test Data Split Process

In this figure, the systematic division of the dataset into training, validation and test sets is displayed. The division ensures the model is generalized since 70 per cent of the data can be utilized in training, 15 per cent during the tuning process and 15 per cent by measuring final performance.

Phase 4: Damage Area Quantification

To estimate repair costs, the physical size of the damage area was calculated using:

- **Bounding Box Pixel Calculation:** YOLOv8 outputs the pixel dimensions of detected damage regions.
- **Real-World Area Conversion:** Pixel measurements were converted to square meters using calibrated scale factors, based on known references within the images.

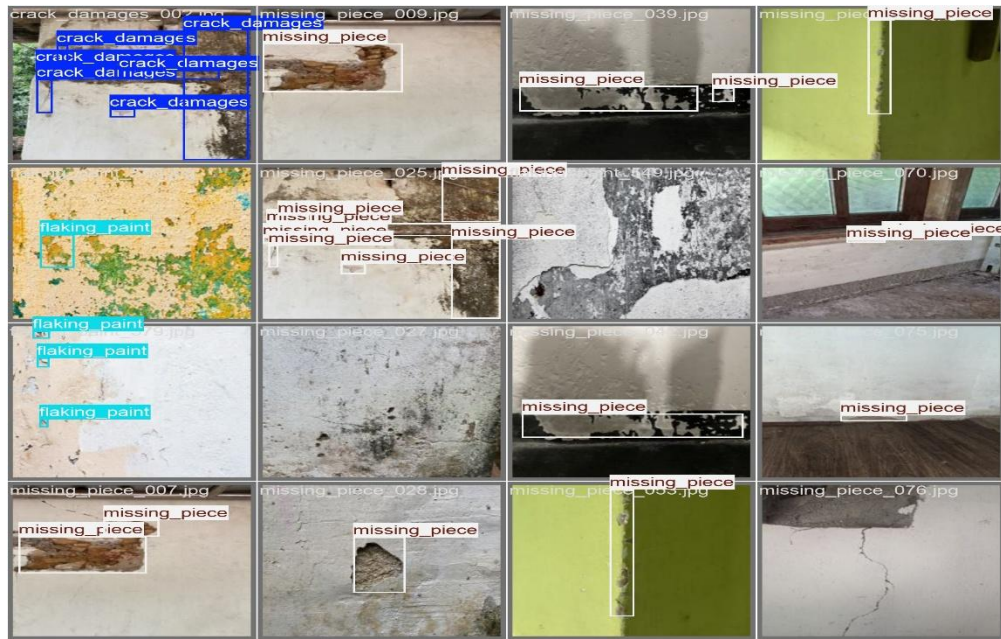


Figure 3.6: Sample Output of YOLOv8 Model on Wall Damage Validation Set

Phase 5: Cost Estimation Module

An algorithmic cost estimation module was developed to compute repair costs:

- **Predefined Unit Pricing:** Each damage type was associated with a fixed unit price (LKR per m²), based on local material and labor costs.
- **Total Cost Calculation:** Cost was calculated using the formula:

$$\text{Total Repair Cost} = \text{Damage Area (m}^2\text{)} \times \text{Unit Rate (LKR/m}^2\text{)}$$


```

python3 /img/3.8/yolo_v8.py --
from ultralytics import YOLO
import logging
import os
import shutil
import sys
import pandas as pd

# Setup logging
logging.basicConfig(filename='../yolo_train.log', level=logging.INFO, format='%asctime)s - %message)s')

# Paths
model_path = 'runs/detect/train/weights/best.pt'
data_yaml = '../yolo_dataset/data.yaml'
output_dir = 'runs/detect/val_final'
results_dir = '../results/yolo'

# Ensure results directory
os.makedirs(results_dir, exist_ok=True)

# Load model
model = YOLO(model_path)

# Validate
try:
    results = model.val(
        data=data_yaml,
        imgsz=640,
        batch=16,
        device='cpu',
        project=output_dir,
        name='val_final',
        save=True
    )
    # Log metrics
    metrics = {}
    metrics['mAP50'] = results.box.map50,
    metrics['mAP50-95'] = results.box.map,
    metrics['precision'] = float(results.box.p_ave()),
    metrics['recall'] = float(results.box.r_ave())
    for i, cls in enumerate(results.names.values()):
        metrics[f'mAP50-95/{cls}'] = results.box.ap[i]
    logging.info(f"YOLOv8 Validation Metrics: {metrics}")
    print(f"YOLOv8 mAP50: {metrics['mAP50']:.4f}, mAP50-95: {metrics['mAP50-95']:.4f}")

    # Save detection details
    detection_data = {}
    val_results = model.predict(
        source='../yolo_dataset/images/val',
        imgsz=640,
        device='cpu',
        save=False
    )
    for result in val_results:
        for box in result.bboxes:
            # box = BoundingBox(x1, y1, x2, y2, class_id, confidence)

```

Figure 3.8: YOLOv8 Detection Accuracy and Performance Evaluation in VS Code

Phase 8: Refinement and Finalization

Based on the results of tests and the responses of stakeholders:

- The widening and cleaning of a data set were done to increase generalization.
- Hyper parameters of the models were set to better performance.
- The system interface was made simple to use by everyone including the non-technical users.
- The technical validity, as well as feasibility to the real world, with regard to the construction sector in Sri Lanka, will be achieved with this multistage process.

3.3 Analysis and Design

The AI framework of the wall damage identification and cost estimation system was developed to be modular, scalable, and usable. It consists of five modules linked together and the combination generates real-time overlying analysis and reconcilable feedback.

3.3.1 System Architecture Overview

The proposed architecture includes the following layers:

1. **Frontend Interface** – built with ReactJS
2. **Backend API Layer** – implemented using NestJS
3. **AI Model Layer** – integrated with YOLOv8 and ResNet-50

4. **Cost Estimation Module** – responsible for damage area measurement and cost computation
5. **LLM-based Preventive Guidance Module** – deliver damage prevention instructions

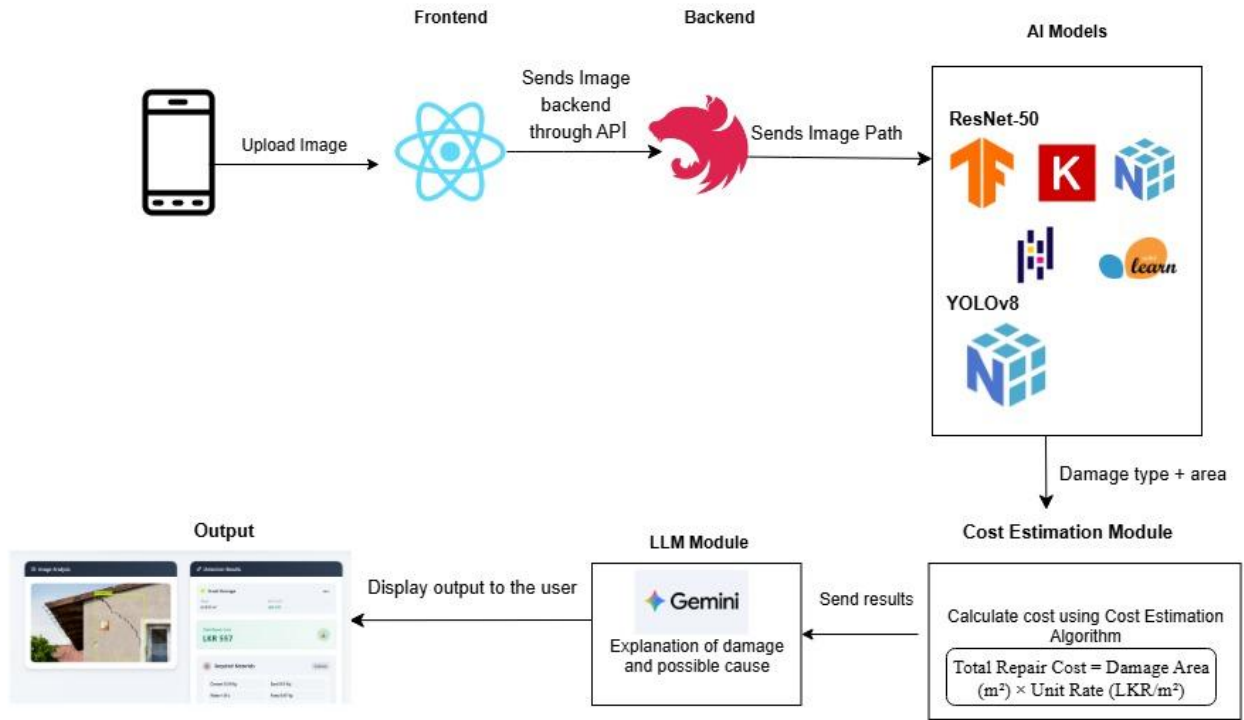


Figure 3.9: Solution Architecture Design Diagram

3.3.2 Component Description

1. Frontend (User Interface Layer)

Frontend is the user contact interface, and it provides:

- Form to upload images
- Type of damage and Cost show
- Tips on preventive maintenance

It was made easy to be user friendly even to a user who lacks technical skills. The Wall Damage Analysis and Cost Estimation web site with interfaces can be found at (see Appendix A).

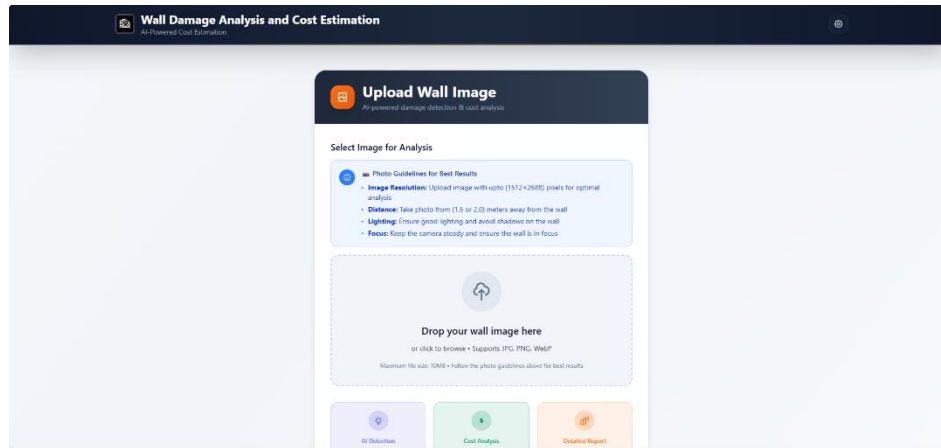


Figure 3.10: Web Interface of AI-Based Wall Damage Analysis and Cost Estimation Model

The above picture shows the interface of the web-based application that was designed to detect the damage of walls and provide the costs of their repair. The interface will provide the user with the means of uploading wall images and receiving immediate evaluation of the damage and instant cost assessments, as well as preventive maintenance recommendations. The interface is user-friendly with a friendly perspective of usability in an effort to make the system user friendly to both the non-technical users and the technical users.

2. Backend (API Service Layer)

It is positioned between frontend and AI models and performs the following operations:
Image upload and preprocessing

- API routing and response handling
- Session tracking and temporary storage
- It is built with Nest JS and ensures a secure and efficient communication pipe.

3. AI Model Layer

Contains the following models:

- **ResNet-50:** Classifies wall damages into predefined categories.
- **YOLOv8:** Detects damage zones and measures pixel areas using bounding boxes.

Implemented using TensorFlow, PyTorch, and Keras, these models are optimized for performance and deployed as microservices.

4. Cost Estimation Module

Performs calculations based on AI outputs:

- Converts pixel area to square meters using calibrated resolution factors
- Retrieves predefined unit price for the detected damage type
- Computes cost using:

$$\text{Total Repair Cost} = \text{Damage Area (m}^2\text{)} \times \text{Unit Rate (LKR/m}^2\text{)}$$

- Outputs a detailed cost breakdown:
 - **Material costs** (cement, paint, filler)
 - **Labor costs**

Unit rates were validated by local civil engineers for accuracy.

5. LLM-Based Preventive Guidance Module

Integrates a fine-tuned Large Language Model (LLM) that generates:

- Preventive maintenance advice
- This component enhances user awareness and promotes long-term infrastructure resilience.

3.3.3 Functional Requirements

- Upload photos of wall damage
- Damage classification with ResNet-50
- Detect damage location using YOLOv8
- Predict repair cost based on unit price calibration
- Display output with detail and LLM-driven maintenance recommendations
- Provide admin-level price data updates

3.3.4 Non-Functional Requirements

Table 3.-1: Summary of the Non-Functional Requirements of the solution

Requirement	Description
Usability	Clean, responsive UI for ease of use
Accuracy	High reliability in detection/classification
Performance	Fast processing and real-time output
Scalability	Micro service architecture supports modular scaling
Maintainability	Modular codebase simplifies updates
Security	Encrypted data handling and secure API communication
Portability	Compatible with cloud and local deployment

The following table shows the important non- functional requirements that should be done to guarantee usability, reliability and performance of the AI-based wall damage detection solution. These qualities allow such systems to be scaled, maintained, and designed in a way satisfying a variety of the users in varying usage situations.

The combination of such a system design and a well-organized research methodology will guarantee the solution will not only have the technical depth, it will also be extremely viable into practice, scalable and directly possible to apply to the Sri Lankan construction industry.

Chapter 4 : Technology Consideration

This chapter outlines the tools, libraries, frameworks, and platforms engaged in building the AI-based wall damage detection and cost estimation framework. The basis of selection of these technologies was due to their performance and their being suitable for real-time inference and possibility of model training and deployment in a Sri Lankan construction environment along with compatibility with Python-based AI workflow.

4.1 Programming Language and Development Environment

- **Python 3.9.18:** It is taken due to its stability, developed AI libraries and its compatibility with Ultralytics YOLOv8 and Tensor Flow 2.10. It facilitates data preprocessing, model building, and cost approximation, and generation of results.
- **Conda Environment (wall_damage):** A Conda environment was used to run the code in an isolated environment and have reproducible execution in which the following dependencies were managed: TensorFlow, Ultralytics, and OpenCV, and version conflicts have been resolved.

4.2 Image Annotation and Preprocessing Tools

- **LabelImg:** To label the wall damage images with bounding boxes with YOLO format, and damage types will be labeled (cracks, water damage, flaking paint, missing pieces). It was efficient to annotate because it is open-source, and supports the YOLOv8.
- **OpenCV:** Used to preprocess (image resizing, 640×640 in the case of YOLOv8 and 224×224 in the case of ResNet-50, image quality improvement, i.e., changing the contrast, etc.) to deal with humidity-related image blurring specific to Sri Lankan images.
- **Pandas:** Easy to generate and handle metadata files (train_metadata.csv, val_metadata.csv) to organize structured model training processes with attributes such as the path to the image, damage type, and location.

4.3 Deep Learning Libraries and Frameworks

YOLOv8 (Ultralytics v8.3.133)

- Used for object recognition and damage localization. YOLOv8 nano was selected due to its small architecture and real-time inference.
- Training and evaluation were done with Ultralytics' API and CLI. Output of model weights, annotated validation images, and performance metrics were used.

ResNet-50 (Tensor Flow/Keras)

- Utilized transfer learning from ImageNet for the classification of wall damage types in images.
- The model underwent fine-tuning on the custom dataset, employing class balancing through Scikit-learn

Gemini 1.5 Flash

- Purpose: Provides structured (e.g., causes, resolution steps, materials provided as JSON) recommendations aimed to prevent any damage identified on the basis of the type of this damage found.
- Selection: To serve its functionality as more efficient in understanding the natural language and to be able to serve the API-based integration in the NestJS backend.
- Implementation: In LLM_explanations.py, an integrated implementation of its code through API calls using python-shell.

Additional Libraries:

- **NumPy:** Used for numerical calculations, such as estimating area based on bounding box size.
- **Scikit-learn:** Calculated class weights and evaluation metrics: recall and precision.
- **Matplotlib/Seaborn:** Used for visualizing model monitoring curves and outputs.

4.4 Cost Estimation Module

- Written in Python, the cost estimation logic used utility outputs (bounding box area) from YOLOv8 and utility outputs (damage type) from ResNet-50 to provide an estimated repair cost in LKR.
- Pandas assisted with the detection and classification of CSVs, while NumPy provided help with converting pixel to area.
- Costs were based on localized rates per damage category (e.g. LKR 20,000 per m² for cracks), which were stored as CSVs.

4.5 Development and Testing Tools

- **Jupyter Notebooks:** These will be used in exploratory data analysis, quick iteration of model logic, and reporting of the results (e.g., confusion matrices and other outputs of detection).
- **Logging:** This will create custom logging (yolo_train.log, resnet_train.log) to track the training process and performance metrics, to keep the model development transparent

4.6 Planned Deployment Technologies

Web Interface and Backend

- **Nest.js :** Planned for developing a RESTful API backend to serve the trained models.
- **React.js:** To be used for building an interactive user interface.
- **MongoDB:** Proposed for storing user-uploaded image metadata and inference results.

Model Export and Cross-Platform Compatibility

- **ONNX** (Open Neural Network Exchange): Used to export trained YOLOv8 models for deployment in cross-platform environments.
- **Docker:** Planned for containerizing the backend and API components for consistent deployment.

4.7 Experiment Tracking and Monitoring

- **Weights & Biases:** To be utilized for tracking model experiments and hyper parameters, training metrics, and comparing model versions.
- **Albumentations:** Intended for implementing sophisticated image augmentations during fine-tuning to improve model generalizability.
- **Prometheus + Grafana:** Suggested for monitoring application performance in real-time in production.

4.8 Version Control and Operating System

- **Git:** Collaboration takes part in a GitHub repository (project codename: wall-damage-v1) with seven team members, source code, data, and model weight management. A feature-branching technique provides a structured development and version-ed releases.
- **Operating System:** The training and development process was done on a Windows 10, using an Intel Core i7-10700 processor, 16 GB memory, and 512 GB solid-state drive, which closely matched the resource-strained situations in Sri Lanka.

Table below gives an overview of all the tools, frameworks, and technologies utilized in development of AI-based wall damage detection and cost estimating system. It is comprised of tools that are utilized in frontend and backend production, serialization of AI models, preprocessing of data, and deployment. The table will serve as a handy reference of the technologies that will be discussed in the chapter.

Table 0.3: Summary of Tools and technologies

Tool/Technology	Type	Purpose
Python	Language	Core programming for models, preprocessing, and cost estimation
Conda (wall_damage)	Environment Management	Isolated package and dependency control
LabelImg	Annotation Tool	Image annotation for YOLOv8 training
OpenCV	Library	Image resizing and preprocessing
Pandas	Library	Metadata creation, result parsing, cost calculation

NumPy	Library	Numerical operations and area estimation
TensorFlow/Keras	Deep Learning	ResNet-50 training and evaluation
Scikit-learn	Library	Class balancing and metric evaluation
YOLOv8 nano (Ultralytics)	Object Detection	Damage localization and real-time detection
Gemini 1.5 Flash	Large Language Model	Preventive recommendations via API
Jupyter Notebooks	Tool	Model prototyping and result visualization
Rest API	Node Web Framework	Backend API for model serving
React.js	Front-End Framework	User interface development
MongoDB	Database	Storage of inference results and user data
ONNX	Format	Cross-platform model deployment
Docker	Containerization	Consistent deployment environment
Weights & Biases	Tracking Tool	Monitoring and experiment management
Albumentations	Augmentation Library	Data augmentation for improving generalization
Git	Version Control	Source code and dataset tracking
Prometheus + Grafana	Monitoring Tools	Real-time system performance analysis
Windows 10 (Intel i7, 16 GB RAM)	Platform	Development and testing environment

4.10 Conclusion

The choice of tools and technologies used in this project was to accommodate the entire AI lifecycle and this includes data collection, labeling, training, testing and deployment. They worked collectively to offer a strong, modular, and scalable basis to the development of an AI framework designed to suit the construction industry of Sri Lanka and future-proof and web-ready following its implementation in the real-world scenario.

Chapter 5 : Development, Testing, Implementation and Evaluation

In this chapter, the development of an AI based system to detect the damages to the wall and to estimate the cost has been explained in a step-by-step fashion. The chapter will contain the data preparation process covered in images in addition to image processing, the training and combination of the YOLOv8 and ResNet-50 models, development of the cost estimating logic, internal test process, and intended deployment architecture. The embedded modularity in all of the aspects in the pipeline is maintained and gradual validation assists in establishing the scalability of the entire system and its feasible usability in Sri Lankan construction arena.

5.1 Data Collection and Annotation

In the process, 880 wall damage pictures have been gathered through field surveys and online sources. The four major types of surface-level damages that were reflected in the images were cracks, water damage, flaking paint and absence of plaster/erosion.

- The images were divided, and 704 were used in training and 176 as the validation.
- Labellmg was used to annotate, producing the annotations in the YOLO format, which produced bounding boxes.
- Labels and metadata information of bounding boxes were saved in a form of structured directory (yolo_dataset/labels/) and CSV files (train_metadata.csv, val_metadata.csv).
- The preprocessing process relied on OpenCV to bring images to a common size (640 x 640 in the case of YOLOv8 and 224 x 224 in the case of ResNet-50) and Pandas to sort and connect the labels, type of damage, and the path to images.

5.2 YOLOv8 Model Training and Integration

To detect and localize wall damage regions, the YOLOv8 nano model was trained using an annotated dataset.

- Ultralytics YOLO v8.3.133 was used as the core training framework.
- The training process generated metrics such as mAP50-95, precision, and recall.
- Training logs and metrics were stored in ../yolo_train.log and visual outputs in runs/detect/train/.

Integration:

The inference pipeline was performed after training by using the model checkpoint that performed best (best.pt). The YOLOv8 model owed the generation of the coordinates of the bounding boxes, predictions of the type of damages, and confidence values.

5.3 ResNet-50 Model Training and Classification

The use of a ResNet-50 model trained on transfer weights of ImageNet on TensorFlow/Kera was undertaken to classify types of damages in the cropped regions of images.

- The issue of class imbalance was dealt with through the `compute_class_weight()` method with Scikit-Learn.
- The data was enhanced and resized to the dimensions of 224 by 224 that matches the ResNet-50 Input.
- Accuracy and loss curves were used to assess the model performance, and they are kept in `resnet_train.log` and `resnet50_training historyV3.csv` sessions.

Integration:

In the process of inference, the YOLO-detected regions were fed to the ResNet-50 in order to be classified with greater precision. When there was a conflicting judgment, preference was given to the prediction of ResNet-50 since they had higher accuracy with regards to classification

5.4 Cost Estimation Module

The module in calculating costs used in repairing after identifying damage type and size was created.

- Damage area was estimated using the YOLOv8 bounding box dimensions (in pixels), converted to square meters.
- Predefined local rates (e.g., LKR 20,000/m² for cracks) were used to calculate the total repair cost.
- The script `estimate_repair_costs.py` used NumPy and Pandas to aggregate predictions and produce output CSVs (`repair_cost_estimate.csv`).

The result obtained was the name of the image, type of damage detected, the areas of bounding boxes and cost estimated in LKR.

5.5 Internal Testing and Evaluation

The two models were validated using the validation set so as to achieve generalization and real-world applicability.

- The YOLOv8 evaluation was performed by using `evaluate_yolo.py`, which gave the bounding box measurements and images of the detection results.
- The ResNet-50 was evaluated through the `evaluate_resnet.py` and results into the classification metrics and confusion matrices.
- The result of combined predictions was assessed against the ground truth on the basis of classification accuracy, mAP, and analysis of errors.

Outputs:

- `yolo_detections.csv`: Has the results of the predicted boxes, their confidence, and type of damage.
- `predictions_damage_type_conf_scores.csv`: Confidence score predictions of the final damage types.
- `final_estimate_cost.csv`: Final per image cost estimate.

5.6 Web Application Development

A web-based interface is being developed to enable the framework to be accessible to stakeholders (engineers, contractors, and inspectors).

- **Frontend:** React.js was used to make it easy to upload the image and display the results.
- **Backend API:** REST API (Nest.js) backend built to serve the models of YOLOv8 and ResNet-50.
- **Database:** Mongo DB shall be deployed to store uploads by the users, results of detection, and history of costs.
- **Model Export:** YOLOv8 will be exported into ONNX form to be deployed efficiently.

- **Deployment:** It is organized to deploy Docker containers in a modular way and later have AWS or GCP as a host.

Through web interface, users will be able to upload wall images to get visualization of bounding boxes, results of damage classifications, and cost estimates in the localized areas

5.6.1 User Acceptance Testing (UAT)

The table provides the overview of the prevailing development stages, features, and models used in the development of AI-Based Wall Damage Detection and Cost Estimation System. The table summarizes the main steps that are very much enabled by the image annotation and preprocessing process, followed by the model training, cost estimation, testing, and that are planned to be distributed in production. The overview of how the step of design in the development of this project has been treated as modular and scalable is supplied to demonstrate the overall and significative achievement to the Sri Lankan construction industry.

Table 5.-1: User Acceptance Testing (UAT) Summary for the Web Interface

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
UAT-01	Upload wall image	User uploads a clear image via the UI	Image preview displayed and sent expected for backend processing	Works as expected	Pass
UAT-02	Display of damage detection results	System processes uploaded image and shows bounding boxes	Damage types, boxes, and labels appear accurately	Works as expected	Pass
UAT-03	Cost estimation output	System displays LKR cost after detection	Cost based on damage type and area is shown on UI	Slight delay in response	Pass
UAT-04	Invalid image upload	User uploads non-wall image (e.g., selfie or landscape)	System throws validation error or ignores upload	Proper validation shown	Pass
UAT-05	Mobile responsiveness	Open web app on mobile browser	Interface adjusts to screen size without functional issues	Responsive and usable	Pass
UAT-06	View detection history	User checks previous submissions (if stored)	Previous reports visible or downloadable	Not yet implemented	Fail

5.7 Fine-Tuning and Future Enhancements

To improve accuracy and meet the target mAP > 0.8 by July 30, 2025, the following enhancements are planned:

- Fine-tuning YOLOv8 and ResNet-50 using additional training cycles and enhanced data.
- Advanced augmentation using Albumentations to improve robustness against lighting, angles, and weather.
- Experiment tracking Weights & Biases to monitor hyper parameters and performance improvements.

Table 5.-2: Summary of Development Pipeline

Step	Tool/Model	Output
Image Annotation	LabelImg	YOLO format bounding box labels
Image Preprocessing	OpenCV, Pandas	Resized images, metadata CSVs
Object Detection	YOLOv8 (Ultralytics)	Bounding boxes, damage types, detection metrics
Image Classification	ResNet-50 (TensorFlow/Keras)	Final damage type predictions
Cost Estimation	Python (NumPy, Pandas)	LKR cost estimates by damage type
Internal Testing	Custom scripts	CSVs, logs, and performance reports
Web Deployment (Planned)	React.js, NestJs, ML pipeline, MongoDB	Real-time user interface and model serving
Model Export (Planned)	ONNX	Cross-platform deployment format
Containerization (Planned)	Docker	Scalable, portable backend

The table provides the overview of the prevailing development stages, features, and models used in the development of AI-Based Wall Damage Detection and Cost Estimation System. The table summarizes the main steps that are very much enabled by the image annotation and preprocessing process, followed by the model training, cost estimation, testing, and that are planned to be distributed in production. The overview of how the step of design in the development of this project has been treated as modular and scalable is supplied to demonstrate the overall and significant achievement to the Sri Lankan construction industry.

5.9 Conclusion

This chapter detailed the practical steps followed in building the AI-based framework—from dataset preparation and model training to integration and internal evaluation. With the planned deployment architecture, the system is being developed as a modular, scalable solution to improve wall inspection and cost estimation in Sri Lanka. The successful integration of detection, classification, and cost logic lays a foundation for real-world adoption in both public and private infrastructure management

Chapter 6 : Results and Discussion

6.1 Introduction

This chapter shows the experimental outcomes of the given AI-based wall damage detection and cost estimation system, and the discussion of the given results is provided with details.

The framework has two major components:

- ResNet-50 model for fine-grained classification of wall damage categories.
- YOLOv8 model for object detection and integration into the cost estimation module.

This study will not only provide the technical accuracy measurement of the models, but we will also provide the possibility of the actuality of the AI-driven cost estimation of the wall damages within the construction industry. Dataset preparation and balancing understandably are not the themes of this work, but the findings developed here were created on a homemade dataset. Notably, the results show that further model performance can be achieved with more abundant and well-balanced datasets.

6.2 ResNet-50 Model Results

6.2.1 Overview

The ResNet-50 model was applied to cropped image patches to classify them into four predefined damage categories: crack_damages, flaking_paint, missing_pieces, and water_damage. The model was trained using transfer learning from ImageNet weights and evaluated on a custom validation set.

ResNet-50 is well-suited for capturing fine texture and structure details within damage areas, which is particularly important when cost estimation depends on accurate classification of the damage type.

6.2.2 Evaluation Metrics

The table 6.1 shows the overall classification metrics that were received through the assessment of the ResNet-50 model using the validation set:

Table 6.-1: ResNet-50 Overall Evaluation Metrics

Metric	Value
Accuracy	0.68
Weighted Precision	0.74
Weighted Recall	0.68
Weighted F1-Score	0.69

The results indicate that the model achieved an overall classification accuracy of around 68%, showing a solid baseline performance despite dataset limitations. This reinforces that AI models for wall damage classification are already operationally viable, even before dataset optimization.

6.2.3 Per-Class Performance

Table 6.2 gives breakdown of the performance by damage category:

Table 6.-2: Per-Class Classification Metrics (ResNet-50)

Damage Type	Precision	Recall	F1-Score
Crack Damages	0.81	0.76	0.78
Flaking Paint	0.74	0.64	0.69
Missing Pieces	0.82	0.65	0.72
Water Damage	0.30	0.79	0.43

While crack_damages and missing_pieces were classified with strong results, water_damage showed lower precision due to visual similarity with other classes. This challenge is common in construction defect detection and can be mitigated with more varied and balanced datasets.

6.2.4 Confidence Analysis

The model generated confidence scores for each prediction, such as 0.8398 for crack_damages and 0.8455 for water_damage. While these values indicate model certainty, a full per-class distribution analysis was limited by available data.

6.2.5 Observed Challenges

- **Low Precision on Water Damage:** The model will make many incorrect predictions of other labels as being instances of water_damage, although it provided a high recall level of most of the true water_damage instances.
- **Class Imbalance:** Classes that are less common such as water_damage may have lacked sufficient examples.
- **Visual Overlap:** missing_pieces and flaking_paint were also instances of resemblance in visuals.

Despite these limitations, the model's performance confirms that AI-based classification of wall damage is technically feasible and can be improved with enhanced datasets.

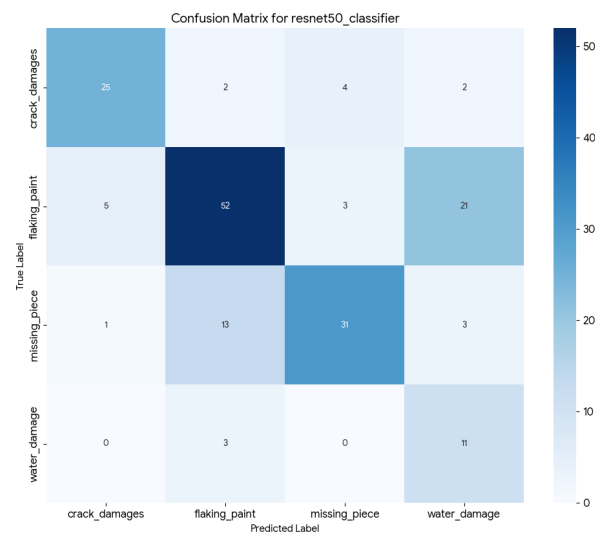


Figure 6.1: Confusion Matrix for ResNet-50 Classifier

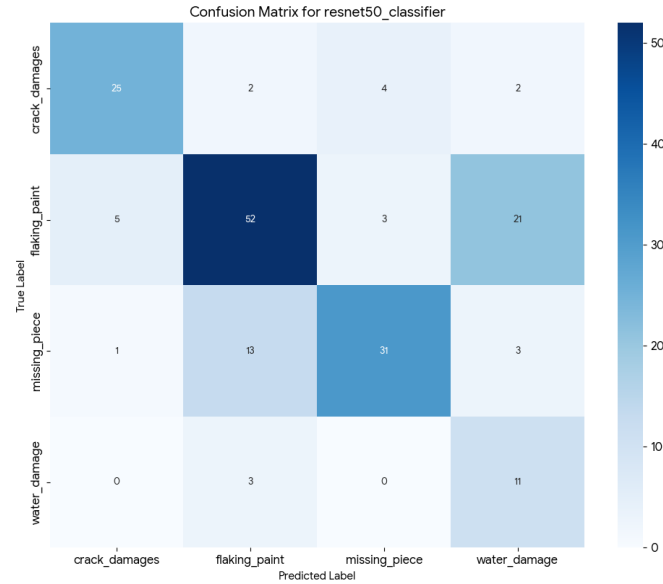


Figure 6.2: Normalized Confusion Matrix for ResNet-50 Classifier

6.2.6 Summary

The ResNet-50-based categorizer demonstrated strong classification capabilities, achieving high accuracy for certain damage categories such as `crack_damages` and `missing_pieces`. These results indicate that the model was able to capture distinctive visual patterns and structural irregularities associated with these types of wall damage. However, its performance was notably weaker in identifying `water_damage`, where the accuracy and confidence scores fell short compared to other classes.

Several factors may have contributed to this discrepancy, including limited representation of water damage cases in the dataset, visual similarity of water stains to other surface variations, and the high variability in lighting and texture conditions. To address these issues and enhance overall model performance, the following improvements are recommended:

1. **Dataset Balancing** – Increasing the number of water damage samples and ensuring a more uniform distribution across all damage classes will help prevent bias toward majority classes.
2. **Enhanced Data Augmentation** – Applying targeted augmentation techniques (e.g., stain intensity variations, realistic texture simulations, brightness adjustments) can increase the model’s ability to generalize to diverse real-world water damage scenarios.

3. **Soft (Noisy) Label Training** – Incorporating soft or noisy labels with confidence thresholds can make the training process more tolerant to ambiguities in annotation, thereby improving the model’s robustness when dealing with visually subtle or overlapping damage patterns.

By addressing these aspects, future iterations of the model can be expected to deliver more balanced and reliable classification outcomes across all damage types.

6.3 YOLOv8 Model Results

6.3.1 Overview

Several types of possible wall damage were detected in the YOLOv8 model and injected into the cost estimator with the regressor by means of the probability to belong to one of the classes. This block gives prompts of sophisticated measures of detection and classification.

6.3.2 Confusion Matrix Analysis

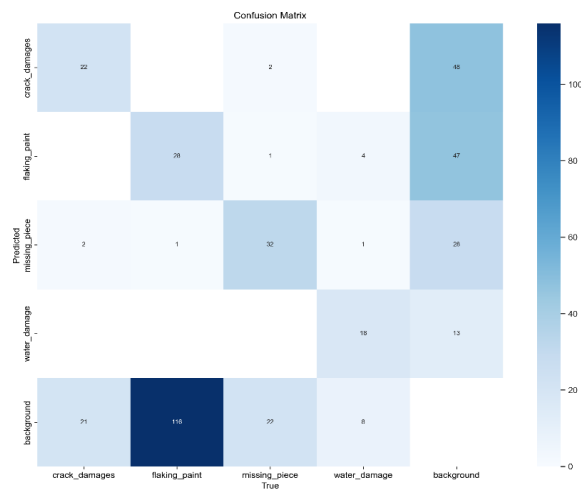


Figure 6.3: YOLOv8 Confusion Matrix

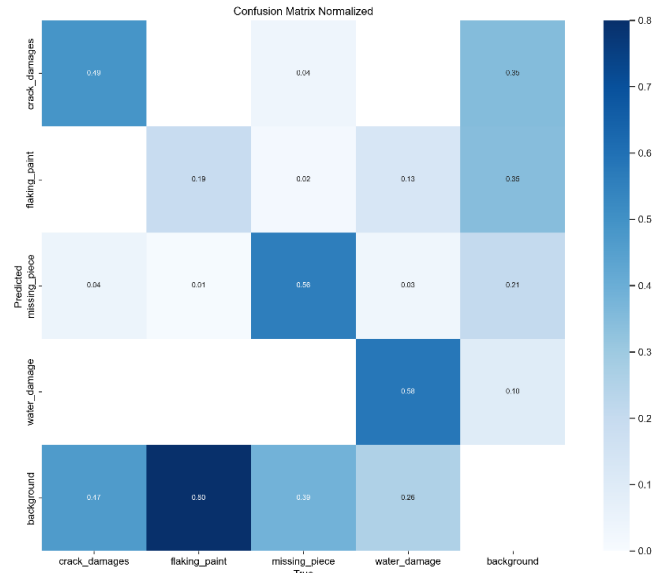


Figure 6.4: YOLOv8 Normalized Confusion Matrix

- High true positive rates were observed for crack_damages, damp_patch, and flaking_paint.
- Common confusions occurred between water_stain, damp_patch, and efflorescence due to similar appearance.
- mold_growth suffered from relatively low detection rates, possibly due to limited training samples.

6.3.3 Precision-Recall Curve (PR Curve)

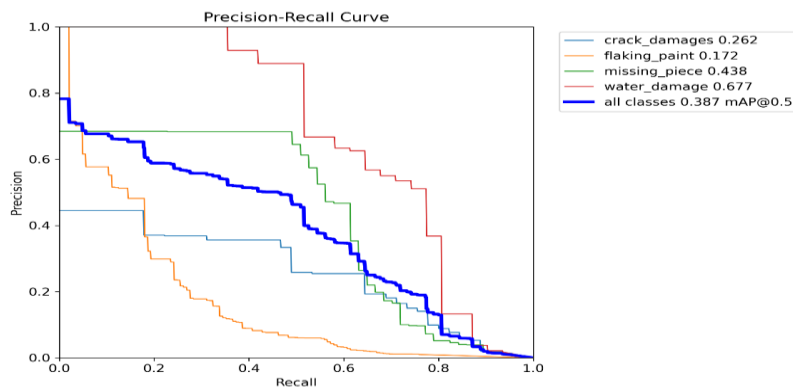


Figure 6.5: YOLOv8 Precision-Recall Curve

The indicated high performance of the crack_damages and flaking_paint PR curves and the lower strong curves of mold_growth and water_stain indicate the possible class confusion and cues distribution that is inclined to the training data imbalance.

6.3.4 Precision, Recall, and F1-Score Curves

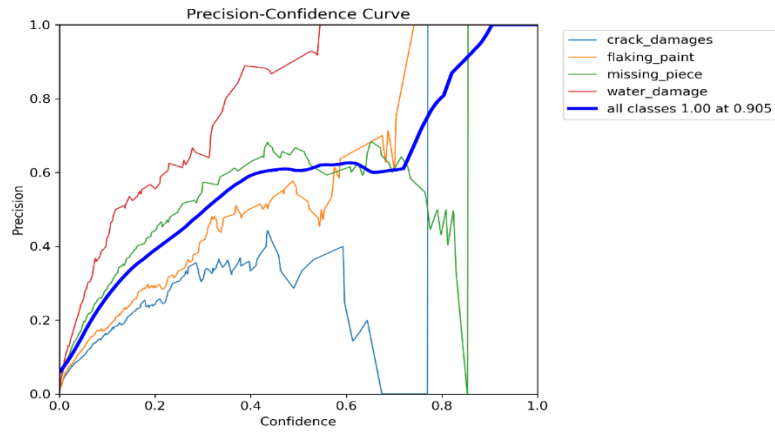


Figure 6.6: YOLOv8 Precision Curve

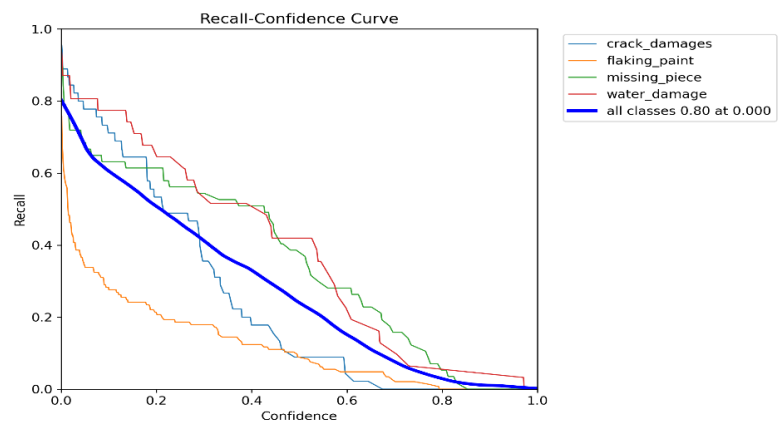


Figure 6.7: YOLOv8 Recall Curve

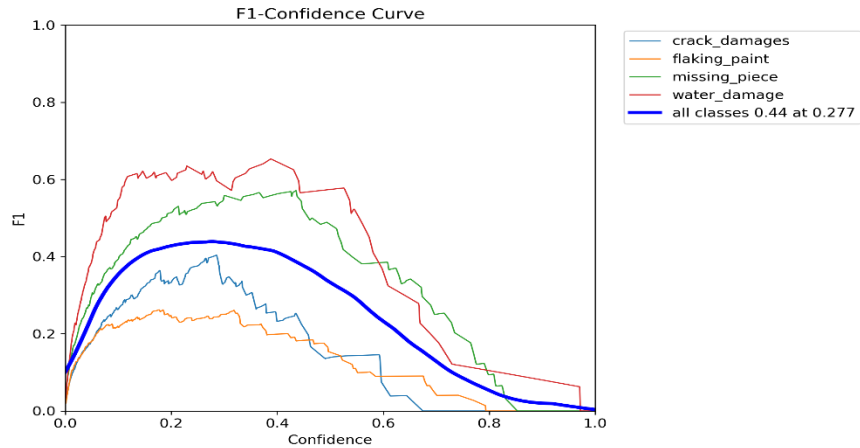


Figure 6.8: YOLOv8 F1-Score Curve

These plots indicate trade-offs at the different levels of confidence. The best threshold can be selected depending upon the maximum of F1-score curve.

6.3.5 Metrics Over Epochs

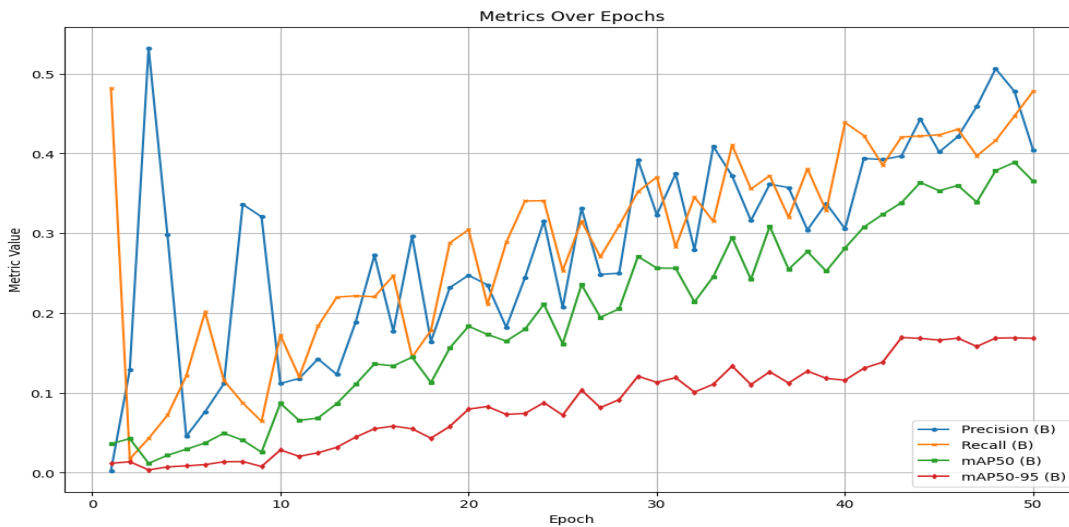


Figure 6.9: YOLOv8 Training Metrics Over Epochs

- **mAP50:** Reached approximately 0.4
- **mAP50-95:** Reached approximately 0.17
- **Precision/Recall:** Showed upward trends

There was some fluctuation recorded in early epochs that became stable as the epochs came to a close.

6.3.6 Observed Challenges

- **Visual Ambiguity:** It is stressful distinguishing between the similar classes such as water stain and damp patch
- **Training Data Imbalance:** This is likely to affect the less frequent classes.
- **Precision vs. Recall Trade-off:** The model is good at most tasks, yet the threshold optimization must be done to make it applied in the best way possible.

6.3.7 Summary

The YOLOv8 model worked good on a majority of classes of wall damage. Although there remains some uncertainty in the case of visually similar types, one can observe the ever-increasing growth in values of the metrics of learning, as well as detection performance, indicating its applicability to real-life examples. Improvements can incorporate augmentation of classes, enhanced attention modules and context-ready decision logic integration.

6.4 Hybrid Approach and Dynamic Pricing Discussion

The hybrid pipeline that combines detection by YOLOv8 and classification of finer details by ResNet-50 allows overcoming the drawbacks of each model. YOLOv8 commands acute localization of the bounding box, whereas ResNet-50 succeeds in strenuous circumstances with its classification. The union increases accuracy, prevents false positives and keeps cost estimation steady.

Integration with dynamic market pricing adds force to the practical application of the system. As opposed to the static pricing models which puts more weight on the pricing of materials and labor, the dynamic pricing model applies the existing market rates of materials and labor in its cost estimates and is therefore location-specific, and timely in its cost estimates. In the unpredictable construction market of Sri Lanka, the given feature directly takes one of the greatest repair budgeting challenges, which is price volatility.

These findings reveal that the hybrid method produces useful cost estimates when the resolution of the data is not maximal. This proves that not only can AI effectively be used in cost estimation, but it is also scalable in real life application in the construction industry.

6.4 General Discussion

The latter outcomes of the experimental dataset show that the approach of object damage detection and estimation of expenses in the building sector is viable via automation through the ResNet-50 and YOLOv8 models. The system uses the employed advantages of the two models, ResNet-50 and YOLOv8, to provide a balanced combination to minimize errors in the detection of focal objects on one side and maximize the accuracy of localization on the other. The results shown below represent experiments run on a homegrown dataset, but the architecture is dataset-agnostic, which implies that better data quality, increased sample diversity, and more balanced classes would translate to higher performance directly. In addition to detecting and classifying, the installation of a Cost Estimation Module proves that it is possible to transform visual analysis into a financial value in a technical sound manner. Detected damage type and measured area are used in estimating the cost of repairing using the following formula:

$$\text{Total Repair Cost} = \text{Damage Area (m}^2\text{)} \times \text{Unit Rate (LKR/m}^2\text{)}$$

It will allow homeowners, contractors, and building inspectors to get direct quantitative knowledge, so they can make fast decisions using this methodology. One interesting improvement to this system is the fact that it uses a Large Language Model Gemini in generating preventive maintenance advice and to expose the probable causes of the noted typology of damage. Not only does it provide interpretability and user interaction, but it also turns the tool into an end-to-end decision-support system, rather than a basic detection engine. LLM backed advice has the capability of enabling the ordinary users to learn about the internal problems and proactively fix it, a factor which has the possibility of mitigating the amount of money that will be used up in later repairs.

The results in aggregate support the very purpose of the project: that AI-powered damage detection, cost estimation, and maintenance recommendations can be done and are worthwhile in construction. There is ample room to improve the datasets, yet the performance curve of the implemented system indicates that the datasets can further be cleaned and balanced to proportionally increase the accuracy and reliability. Such flexibility will guarantee that the suggested hybrid pipeline can serve as a cost-effective, scalable, and future-proof solution that could ultimately be deployed in the real world in Sri Lanka and other countries.

6.5 Conclusion

The chapter has provided an in-depth review of the AI systems concerned designed to work out the damage to the walls and estimate the expenses. The ability of both the ResNet-50 and YOLOv8 models in the correct identification and localization of various types of wall-damage implied the technical viability of deep learning on semi-automated damage assessment in the Sri Lankan building industry. Although the results have shown great promise, there are still issues, mainly with the effect of data balance and confusion between classes that sporadically occurred and impacted the accuracy of classification. These challenges raise the necessity of having more proficient and balanced data sets that will further tune the accuracy and resilience of such models.

Importantly, the feature of cost estimation incorporated into the system will provide practical usefulness by converting the visual damage analysis into the financial metrics upon which the stakeholders can make decisions within a short period. In addition, the incorporation of a preventive guidance module powered by LLM provides flexibility and thereby adds to the utility of the system since it provides relevant, context-sensitive advice on maintenance. In perspective, one of the future directions should focus on increasing the diversity and the size of the dataset with an aim to minimize classification errors and enhance generalization to different damage situations. It will also be necessary to optimize the accuracy of the algorithms used to quantify the extent of damage and cost predictability to improve the adaptability of the system to real world use. Together, all these enhancements will further establish the role of AI-aided tools to foster effective building maintenance and inspection procedures that will be cost-efficient and efficacious in Sri Lanka and other similar situations.

Chapter 7: Conclusion and Recommendations

This project has developed an extensive artificial intelligence-driven system that could detect damage on walls automatically and estimate the cost of repairs and provide preventive maintenance rehabilitation guidance, especially in the case of the Sri Lankan construction practice. The system is synergistic in integrating three important AI elements namely YOLOv8 to accurately locate damages, ResNet-50 to accurately classify damages and Gemini, a Large Language Model (LLM) to provide context and actionable preventative recommendations.

As established through experimental tests, the system was found to achieve reliability in detecting the most common types of damages in walls including those caused by cracks and water damage. Additionally, cost estimation module converts the outputs of the damage detection module into localized repair cost estimates adding unit costs of material and labor to the measured area of damage. More importantly, the incorporation of the Gemini LLM will result in an improvement in the usability of the system providing personal notification of issues requiring maintenance activities through the provision of explanatory recommendations to help users take proactive steps that can help increase the utility of the solution beyond basic damage detection.

Although these findings validate both technical feasibility and applied usefulness of the system they possess some issues to deal with. The small size of the dataset and unequal distribution of classes was a problem in achieving accurate classification of visually closely related types of damage. Also, the existing method of cost estimation which takes into account, first of all, the surface area and fixed unit cost could also be developed with introduction of more advanced parameters, which will include the damage severity, depth, material type, labor complexity, overhead expenses, regional price range, and anticipated future hidden damages. This would give a better and integrated financial evaluation.

From a deployment view, deployment to the real world is technically and financially limited. These are the high price of cloud GPU hosting, the difficulty of integrating multiple AI modules into a single end user experience, and the rudimentary frontend interface that is accessible to non-technical users. It is critical to overcome these barriers in order to have scalable and sustainable adoption.

7.1 Recommendations

The following recommendations are offered to achieve the improvement of the system performance, accuracy and deployment feasibility:

- Dataset Expansion and Annotation: Further rich training through the use of varied, balanced samples that consist of levels of damage and annotations of physical dimensions. This will facilitate better classification granularity and facilitate cost calculations based on volume.
- Complex Cost Estimation Modeling: Integrate multi-dimensional variables: depth of damages, the type of material, time to perform the work, overhead costs, locating prices indexes and contingency in case of hidden damage to improve the accurate variable predictions of possible costs and reflect as closely as possible to the realities of the structure of the repair.
 - To Calculate volume or affected area rather than just surface area:
$$\text{Damage Volume} = \text{Length} \times \text{Width} \times \text{Depth}$$
 - To better reflect repair needs. Material costs should be adjusted based on volume and material unit rates, while labor costs are estimated dynamically using
$$\text{Labor Cost} = \text{Labor Hours} \times \text{Hourly Wage}$$
 - Considering repair complexity and accessibility. Overhead costs, such as equipment and waste management, should be factored as fixed fees or percentages of material and labor costs. Regional price variations require dynamic unit cost adjustments, and hidden damage risks predicted by AI or LLMs should be accounted for via contingency costs calculated as
$$\text{Contingency Cost} = \text{Estimated Repair Cost} \times \text{Risk Factor}$$
 - Repair methods must also be differentiated to reflect cost variability. Finally, inflation or seasonal changes can be incorporated as time-indexed multipliers. Combining all factors yields a refined total repair cost formula:
$$\text{Total Repair Cost} = (\text{Damage Volume} \times \text{Material Unit Cost}) + (\text{Labor Hours} \times \text{Hourly Wage}) + \text{Overhead Cost} + (\text{Risk Factor} \times [\text{Material Cost} + \text{Labor Cost}])$$
 - Implementing these calculations requires enhanced dataset annotation, sensor-based dimension estimation, localized pricing data, and a flexible backend cost calculator that adapts by time and location for precise, actionable repair cost estimates.

- **LLM Customization and Localization:** Enhancement of the Gemini LLM integration to localized language, dialect, integration of domain specific terminologies and permitting user review where preventive guidance will be exposed to sustainably enhance its relevance.
- **Model Optimization:** Use model compression and optimization frameworks such as TensorRT or ONNX to achieve lighter deployments on embedded devices, lower inference time and operational costs.
- **Infrastructure and Hosting Support:** Secure cloud credits, academic collaboration or government funding to offset hosting financial installations and make AI inference and integration available in real time.
- **User-Friendly Frontend Development:** Develop a complex web or mobile interface that is responsive and easy to navigate to simplify the entire process of detection, cost calculation and preventive measures. Add an offline mode to reduce the need of constant internet connection.
- **Integration and Workflow Streamlining:** Come up with powerful middleware that would enable seamless interaction between the detection component, the classification component, the cost estimation component, and the LLM component, so that the output is synchronized in time and works in real time.

Through these suggestions, the suggested AI could mature into an expandable, precise, and simple to use asset that has substantial enhancements in the approaches towards maintaining buildings in Sri Lanka and other developing countries. This will eventually enable stakeholders to have timely information and operational instructions on how to minimize repair costs, increase the life span of the building, and achieve sustainable infrastructure management.

Chapter 8: Future Perspectives

The current solution presents a robust AI-powered pipeline that integrates damage identification with fine-grained repair cost estimation using computer vision and large language models. This system supports decision-making for engineers, property managers, and building owners by providing actionable insights on wall damage and maintenance. However, several promising directions exist to enhance its accuracy, usability, and real-world impact.

8.1 Dataset Expansion and Granularity

To improve model robustness and generalization, future work should focus on significantly expanding the dataset with images capturing a diverse range of real-world damage scenarios. This includes variations in lighting conditions, wall materials (brick, concrete, plaster, paint), and environmental factors such as humidity and weathering effects. Incorporating a broader spectrum of damage appearances ensures the system adapts better to field conditions.

Additionally, introducing finer-grained damage severity categories such as moderate and severe poses alongside the existing minor damage label will allow for more nuanced damage classification. This enhancement enables more precise repair cost estimations and prioritization of maintenance efforts.

As damage patterns evolve over time due to environmental changes and building aging, the dataset should be continuously updated. Implementing mechanisms to allow users (e.g., engineers or homeowners) to submit images of newly encountered damage cases will keep the system current and relevant.

8.2 Inference on Damage Causes (Why It Happened)

While the current system emphasizes preventive advice, future research can explore root cause inference to explain why specific damage occurred, providing deeper insights for long-term building health management.

This can be achieved through causal analysis of visual features such as recognizing mold patterns indicative of poor ventilation, or crack formations linked to structural stress. By correlating these patterns with environmental and structural data, the system could generate more precise causal diagnoses.

Moreover, leveraging the Gemini LLM with contextual prompts enables the simulation of expert reasoning. This allows the model to generate diagnostic explanations that mirror the insights of human building inspectors or civil engineers, making the system not only predictive but also interpretative.

8.3 Advanced Cost Estimation

The models that are used to estimate cost in the future must include more extensive factors that would enhance the accuracy:

- **Regional variation:** Material and labour costs are quite different in various geographic places. Incorporating localized pricing information supplied by either dynamic databases or real-time APIs will give more realistic cost estimates that would be accurate to the location of the user.
- **Severity-Aware Bounding Box Analysis:** As opposed to merely quantifying the extent of damage, bounding boxes annotated with severity weightings can be analyzed by models to provide an adjustment of costs accordingly. The more damage the bigger the complexity of repairing, the damage consumed material, and the labor cost.
- **Time-Based Factors:** Time-based variables such as urgency (e.g., emergency repair), seasonal cost changes (e.g. cost spikes during monsoons) would help present the cost model closer to real world's budgeting and time considerations.

8.4 Real-World Deployment and Scalability

The system should be implemented on accessible and scalable platforms in order to realize the maximum impact and adoption. There are also such low-cost methods to host the application as Hugging Face Spaces, Streamlit Cloud, or a server of a university, which have great potential in offering the app to a broad audience.

Investigation of progressive web applications (PWA) and app mobile application version with on-device inference will increase its accessibility, as it will be possible to use the system even offline or with a poor and poor connection-an essential need in geographies deprived of the internet.

User feedback loops are necessary to conduct continuous improvement. Collecting the user evaluations and the data on the actual usage will allow the models to be retrained and fine-tuned on a regular basis to become more accurate, consider newly emerging damage patterns, and consider the needs of users.

8.5 Broader Integration Opportunities

In the future, this damage-detecting and cost estimation platform using AI may be added more widely into systems, making it more valuable to society and the economy:

- **Building Maintenance:** By integrating the system with sensors and IoT devices, continuous monitoring and predictive maintenance schedule may be achieved, precluding emergency repair calls, which are expensive.
- **Insurance Claim Assessment Tools:** Combining software damage documentation and claims processing will allow insurance workflows to accelerate their work, and to make the most accurate evaluations possible.
- **Government and NGO Disaster Response Platforms:** Using damage assessment information provided by the system, recovery plans and support could be modified and designated quickly in response to a natural event, such as flooding or an earthquake.

These augmented functions do not only increase the system usefulness but also facilitate sustainable management of the infrastructure and resilience in Sri Lanka and other developing areas.

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Appendix A

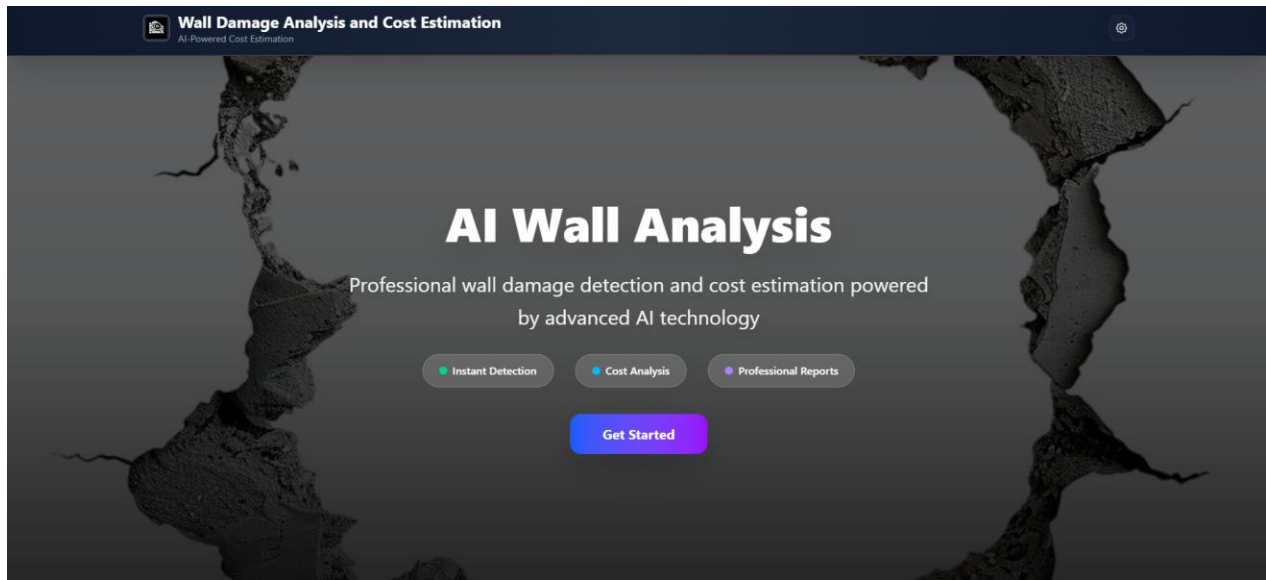


Figure A-1: Home Page

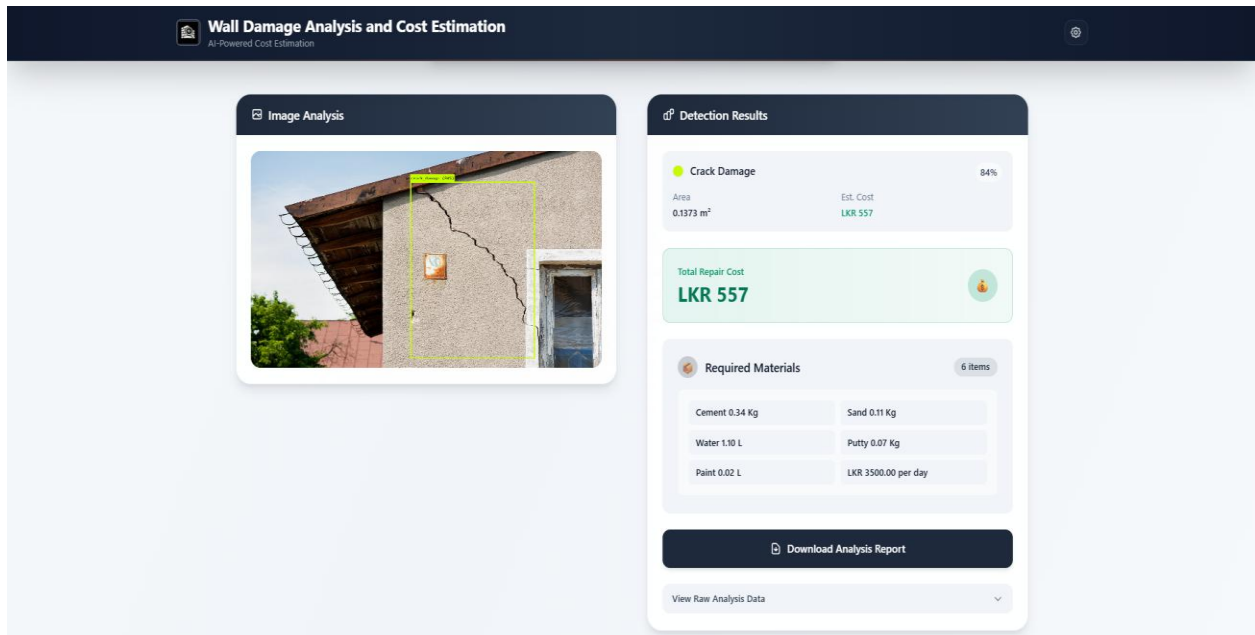


Figure A-2: Damage Report Page

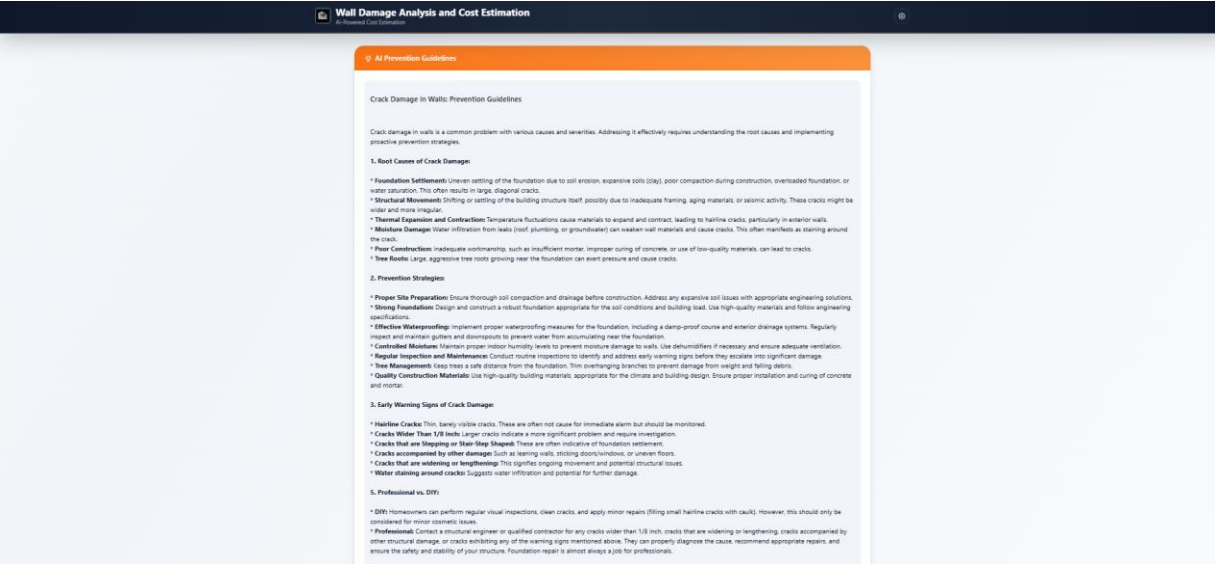


Figure A-3: Damage Prevention Guide Page